

## China CDMO

# China CDMO: Primer on Peptide CDMOs Part 1 - Supply-side leaders differentiate in scale, cost, and technology breadth



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*This report is the first in a multi-part primer series on CDMOs. We start with a supply-side review of the peptide industry. Future reports will examine GLP-1-driven demand growth, peptide market sizing, and the long-term supply-demand balance.*

**A fragmented industry increasingly dominated by scaled platforms.** The peptide CDMO landscape spans three groups: diversified CRDMOs - Thermo Fisher and WuXi AppTec among others, specialty peptide leaders - Bachem, PolyPeptide, CordenPharma, and a long tail of emerging suppliers. Historically, peptide outsourcing was led by specialists with deep manufacturing know-how, but the rapid growth of metabolic peptides has increasingly favored suppliers able to deploy capital and capacity at scale. **Chinese CRDMOs have been the primary beneficiaries of the GLP-1 boom.** Revenue of the top four peptide CDMOs expanded from c. **\$1 Bn in 2021 to \$3 Bn in 2025**. WuXi AppTec's TIDES business grew from **\$120 Mn to \$1.7Bn**, increasing market share from **11% to 54%**, while Asymchem's peptide business accelerated to **124% growth in 2025** from a much smaller base. By comparison, Bachem and PolyPeptide delivered more modest growth of mid single digit to teens, respectively, as market growth became increasingly concentrated in large-scale metabolic peptide programs.

**Cost structures differ dramatically.** There is a wide range of profitability across peptide CDMOs despite operating in the same value chain. WuXi AppTec generates industry-leading margins with approximately **47% gross margin and 37% EBIT margin**, compared with roughly **22% EBIT margin at Bachem** (already leading among Western peptide players), while PolyPeptide and EuroAPI remain near breakeven. **A major factor of the diverse economics is scale.** Capacity disclosures are scarce, and we prefer peptide output in metric tons over reactor volume as a measure of manufacturing scale. WuXi AppTec remains the clear industry leader with an estimated **58 metric tons of peptide output in 2025**, but beneath it the industry is dispersed across multiple tiers: **Corden and Asymchem occupy the 10-20 ton range**, while **Bachem and PolyPeptide and other Chinese peptide specialties** are generally in the **1-10 ton range**. A long tail of regional suppliers remains below 1 ton of annual capacity.

**Technology and efficiency are further differentiators beyond capacity.** Basic peptide capabilities such as cyclic and stapled peptides are now broadly available across most leading suppliers. Differentiation increasingly comes from **conjugation technologies, sustainability, and speed of execution**. WuXi AppTec and Asymchem offer the broadest technology portfolios across **peptide-oligonucleotide, radionuclide and other advanced conjugates**, while WuXi and Bachem appear to lead in process efficiency and waste reduction. Chinese players also generally deliver shorter peptide IND-to-NDA timelines than Western peers, strengthening their competitive position as peptide innovation expands into increasingly complex modalities.

## BERNSTEIN TICKER TABLE

| Ticker                   | Rating | 22 Jul 2026 |                  | Price<br>Target | TTM<br>Rel.<br>Perf. | Reported EPS |       |       |       | Reported P/E (x) |       |       |
|--------------------------|--------|-------------|------------------|-----------------|----------------------|--------------|-------|-------|-------|------------------|-------|-------|
|                          |        | Cur         | Closing<br>Price |                 |                      | Cur          | 2025A | 2026E | 2027E | 2025A            | 2026E | 2027E |
| 002821.CH (Asymchem)     | O      | CNY         | 164.28           | 198.00          | 29.7%                | CNY          | 3.16  | 4.04  | 6.07  | 22.5%            | 38.6% | 56.5% |
| 603259.CH (WuXi AppTec)  | O      | CNY         | 125.90           | 168.00          | 19.8%                | CNY          | 6.72  | 6.36  | 7.37  | 40.1%            | 4.8%  | 16.9% |
| 2269.HK (WuXi Biologics) | M      | HKD         | 38.84            | 42.00           | 5.2%                 | CNY          | 1.22  | 1.54  | 1.85  | 31.8             | 25.3  | 21.0  |
| 6821.HK (Asymchem)       | O      | HKD         | 118.90           | 157.00          | (7.4)%               | CNY          | 3.16  | 4.04  | 6.07  | 32.5             | 25.4  | 16.9  |
| 2359.HK (WuXi AppTec)    | O      | HKD         | 156.70           | 204.00          | 38.3%                | CNY          | 6.72  | 6.36  | 7.37  | 20.1             | 21.3  | 18.3  |
| ASIAx                    |        |             | 1,909.09         |                 |                      |              |       |       |       |                  |       |       |

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

002821.CH, 603259.CH valuation is EPS Reported CAGR;

Source: Bloomberg, Bernstein estimates and analysis.

## INVESTMENT IMPLICATIONS

We rate Wuxi AppTec and Asymchem Outperform; Wuxi Biologics Market-Perform.

## DETAILS

### PEPTIDE CDMOS OVERVIEW

#### Lay of the land: a fragmented industry anchored by a few of scaled leaders

The peptide CDMO industry remains highly fragmented, but participants can broadly be grouped into three categories based on scale and strategic focus. The first group comprises **large diversified CRDMOs** with enterprise revenues exceeding \$1 Bn, including **Thermo Fisher** (covered by Eve Burstein), **WuXi AppTec**, **EuroAPI** (covered by Justin Smith), and **Asymchem**. While peptides typically represent only a portion of their broader service portfolio, these companies benefit from global customer access, integrated development-to-commercialization platforms and substantial capital resources.

Then there are the **specialty peptide CDMOs**, predominantly focused on peptides and related modalities like oligonucleotides. Key players include Bachem (not covered), PolyPeptide Group (not covered) and CordenPharma (private, not covered), each generating peptide-related revenues in the \$0.1-1.0 Bn range. They differentiate with long histories, peptide-specific know-how and regulatory track records, making them preferred partners for innovator companies. Historically, they have dominated peptide outsourcing through specialized manufacturing expertise. However, the rapid expansion of metabolic peptides has increasingly attracted competition from larger CRDMOs able to deploy capital at scale.

Finally, **smaller CDMOs** and pharma/biotech companies with peptide manufacturing operations include a long tail of companies. A few in China (Jiuzhou Pharma, Porton, and Apeloa, all not covered) are expanding their peptide business in API and intermediate CDMO and loading up capacities rapidly (Exhibit 1).

#### Revenue and market share evolution: the rise of Wuxi Aptec (and Asymchem next)

The competitive landscape has shifted dramatically over the past five years. Among leading peptide CDMOs, combined revenue expanded from approximately **\$1 Bn in 2021 to \$3 Bn in 2025**, driven primarily by demand for GLP-1 and other metabolic peptides. WuXi AppTec has been the largest beneficiary, with TIDES revenue increasing from c.**\$120 Mn in 2021 to \$1.7 Bn in 2025 with over 70% compound growth**, driving market share from **11% to 54% (Exhibit 4)**. Early and aggressive capacity investments made during the early 2020s enabled the company to capture a disproportionate share of industry growth.

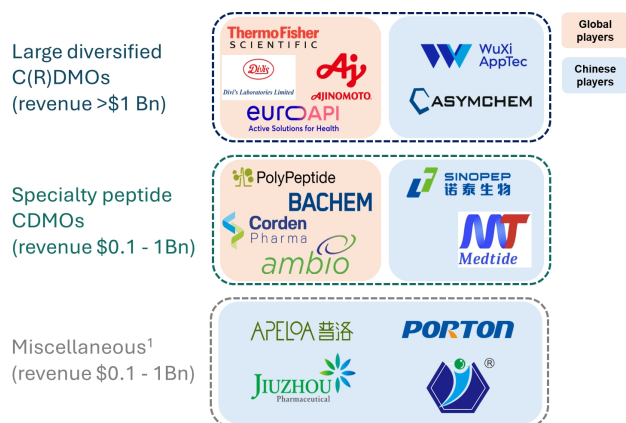
In contrast, traditional peptide specialists have seen more modest growth. Bachem's peptide revenue increased from \$716 Mn in 2023 to \$862 Mn in 2025, while PolyPeptide grew from \$365 Mn to \$444 Mn, both growing at **single digit to mid-teens**. Although both remain leading suppliers with strong technical capabilities, their market shares have gradually diluted as industry growth became increasingly concentrated in large-scale metabolic peptides. Meanwhile, **Asymchem remains small at \$152 Mn revenue in 2025 but is entering a rapid growth phase**, with revenue growth accelerating to **124% in 2025 (Exhibit 5)**. Asymchem is another potential future share gainer as newly built peptide capacity ramps. Overall, recent industry growth has been led overwhelmingly by Chinese CRDMOs, while Western specialty peptide CDMOs have grown more slowly despite continued expansion.

## EXHIBIT 1: Financial and operating metrics of global leading peptide CDMOs

| Company name                           | Company type               | Financial metrics                           |                    |                     |                                    | Operating metrics                         |                                              |                       |
|----------------------------------------|----------------------------|---------------------------------------------|--------------------|---------------------|------------------------------------|-------------------------------------------|----------------------------------------------|-----------------------|
|                                        |                            | 2025 total revenue (\$ Mn)                  | 2025 CAPEX / Sales | 2025 ROIC           | 2026 revenue growth <sup>3</sup> % | Latest peptide capacity                   | # of peptide projects 2025                   | Core synthesis method |
| WuXi AppTec                            | Large diversified C(R)DMOs | 6,818 (group); 1,705 (TIDES)                | 12%                | 38%                 | 18% - 22% (group)<br>40% (TIDES)   | 100kL SPPS; est. 58 tons produced in 2025 | 472 (D&M)                                    | SPPS                  |
| ASYMCHEM                               | Large diversified C(R)DMOs | 1,001 (group); 154 (peptide)                | 19%                | 10%                 | 19% - 22% (group)                  | 45kL SPPS; 69kL by 2026                   | 52                                           | SPPS                  |
| Corden Pharma                          | Large diversified C(R)DMOs | 1,094 (group); c.600 <sup>2</sup> (peptide) |                    |                     |                                    | >17kL SPPS; >42kL targets by 2028         |                                              | SPPS + LPPS + Hybrid  |
| euroAPI<br>Active Solutions for Health | Large diversified C(R)DMOs | 972 (group); 71 (large molecule)            | 9%                 | 0%                  | 10% decline                        | c.500kg in 2025 at Frankfurt site         | c.20-30 large molecule projects <sup>4</sup> | SPPS                  |
| BACHEM                                 | Specialized TIDES CDMOs    | 862                                         | 47%                | 6%                  | 35% - 45%                          | 6 peptide manufacture sites globally      | 152 (Chemistry, Manufacturing and Controls)  | SPPS                  |
| PolyPeptide GROUP                      | Specialized TIDES CDMOs    | 447 (group); 252 (metabolics)               | 28%                | 1%                  | 20% - 25%                          | 6 peptide manufacture sites globally      | 47 (37 clinical + 10 commercial)             | SPPS                  |
| MT                                     | Specialized TIDES CRDMOs   | 86                                          | 23%                | 27.8% (2025 HK IPO) |                                    | >1 tons API produced                      | 1,614 (328 CDMO, 9 NCE GLP-1s)               | SPPS + LPPS + Hybrid  |
| SINOPEP<br>诺泰生物                        | Miscellaneous <sup>1</sup> | 291 (group); 72 (CDMO)                      | 70%                | 5.5%                |                                    | 64.7kL SPPS; 6.2 tons in 2Q25             |                                              | SPPS                  |
| 圣诺生物                                   | Miscellaneous <sup>1</sup> | 111 (group); 17 (CDMO)                      | 26%                | 8.8%                |                                    | 2.5 tons in 4Q25; targeted 10 tons        | 40+ (20+ clinical + 3 commercial)            | SPPS                  |

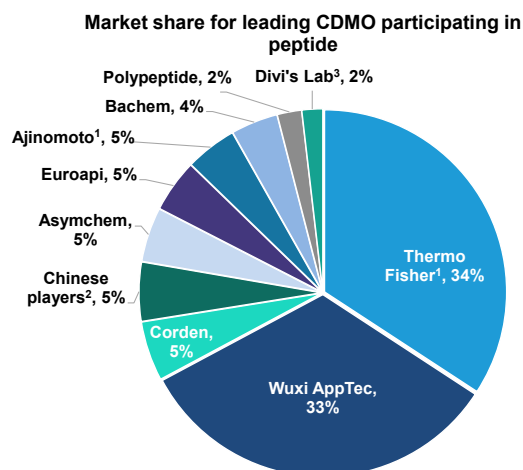
Note: 1. Miscellaneous represent smaller CDMOs or pharma/biotech with peptide CDMO segment; 2. BernE assuming peptide business account for 60% of total revenue for Corden; 3. Company guidance; 4. BernE based on its 2022 disclosure  
Source: Company disclosures, Bernstein analysis and estimates

## EXHIBIT 2: Global peptide CDMO market landscape by business focus



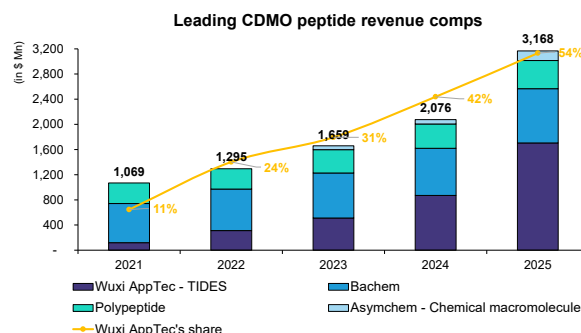
Note: 1. Represent smaller CDMOs or pharma/biotech with peptide CDMO segment; Ambio is acquired by Corden in 2026  
Source: Company disclosures, Bernstein analysis

## EXHIBIT 3: Thermo Fisher and WuXi AppTec anchor the peptide-participating CDMO landscape, together commanding two-thirds of the market



Note: 1. Revenue from CDMO business; 2. Chinese players include SinoPep, snBiopharma, Medtime, Jiuzhou Pharma, and Apelo; 3. Divi's CDMO revenue estimated assuming CDMO contributes 35–40% of total revenue.  
Source: Company disclosures, Bernstein analysis and estimates

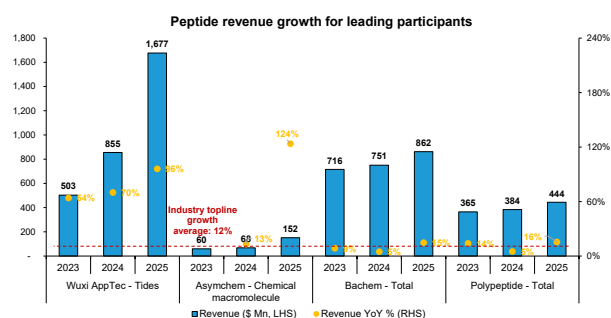
#### EXHIBIT 4: Revenue trends indicate sustained expansion among leading peptide participants, with Wuxi AppTec demonstrating the strongest growth trajectory



Note: \* Total company revenue used where peptide revenue was not separately disclosed

Source: Company disclosures, Bernstein analysis

#### EXHIBIT 5: WuXi AppTec and Asymchem significantly outpace global peers, delivering rapid peptide revenue growth driven by capacity expansion



Source: Company disclosures, Bernstein analysis

### PLAYER BENCHMARKS: SCALE, COST STRUCTURE, AND TECHNOLOGY

#### Scale: despite limited disclosures in the industry, Wuxi Apptec is on a solid level of its own

Assessing peptide manufacturing scale remains challenging due to limited public disclosure. Among major peptide CDMOs, only **WuXi AppTec and Asymchem have disclosed aggregate peptide capacity in kiloliters (kLs)**, reporting **c.100kL and 45kL in 2025**, respectively, with planned expansion to **130kL and 69kL in 2026**. For most other players, capacity must be inferred from individual site disclosures, announced expansions, revenue scale and industry checks. More importantly, we believe **metric tonnes of peptide output are a better measure than kL of reactor volume**, as productivity is heavily influenced by peptide length, loading, yields and batch configuration. On this basis, WuXi AppTec stands in a category of its own, with estimated output of **~58 metric tons in 2025**, while most peers remain in the range of **single-digit to low-double-digit metric tons**, highlighting the substantial scale advantage WuXi has established during the GLP-1 expansion cycle (Exhibit 6).

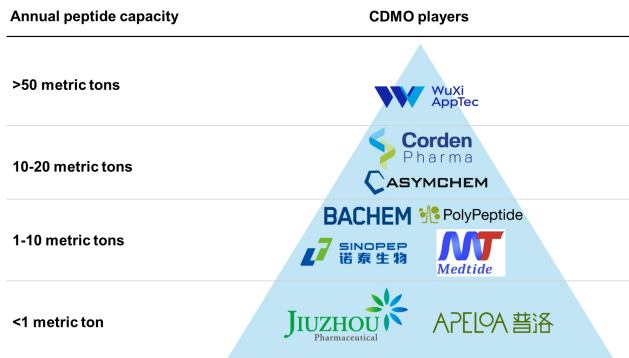
#### Large batch capability remains a differentiator

Reported commercial batch sizes vary significantly across suppliers, reflecting differences in technology platforms, peptide complexity and customer mix. Most leading peptide CDMOs operate at average commercial batch sizes of roughly **15-30kg**, with only a handful of players possessing demonstrated track records in consistently executing larger-scale campaigns (Exhibit 7).

#### Chinese capacity expansion has yet to be globalized

Chinese players, led by **WuXi AppTec and Asymchem**, have expanded peptide capacity aggressively and now rank among the largest suppliers globally. However, unlike Western peers such as **Bachem, PolyPeptide and Corden**, whose manufacturing networks are distributed across North America and Europe, peptide production capacity at Chinese leaders remains concentrated almost entirely within China today. While WuXi plans to add peptide manufacturing capacity in Singapore by 2028, geographic diversification remains limited at present, leaving Chinese suppliers more exposed to customer concerns around supply chain concentration and geopolitical risk (Exhibit 8).

EXHIBIT 6: Global peptide CDMO capacity comparison (in metric tons)



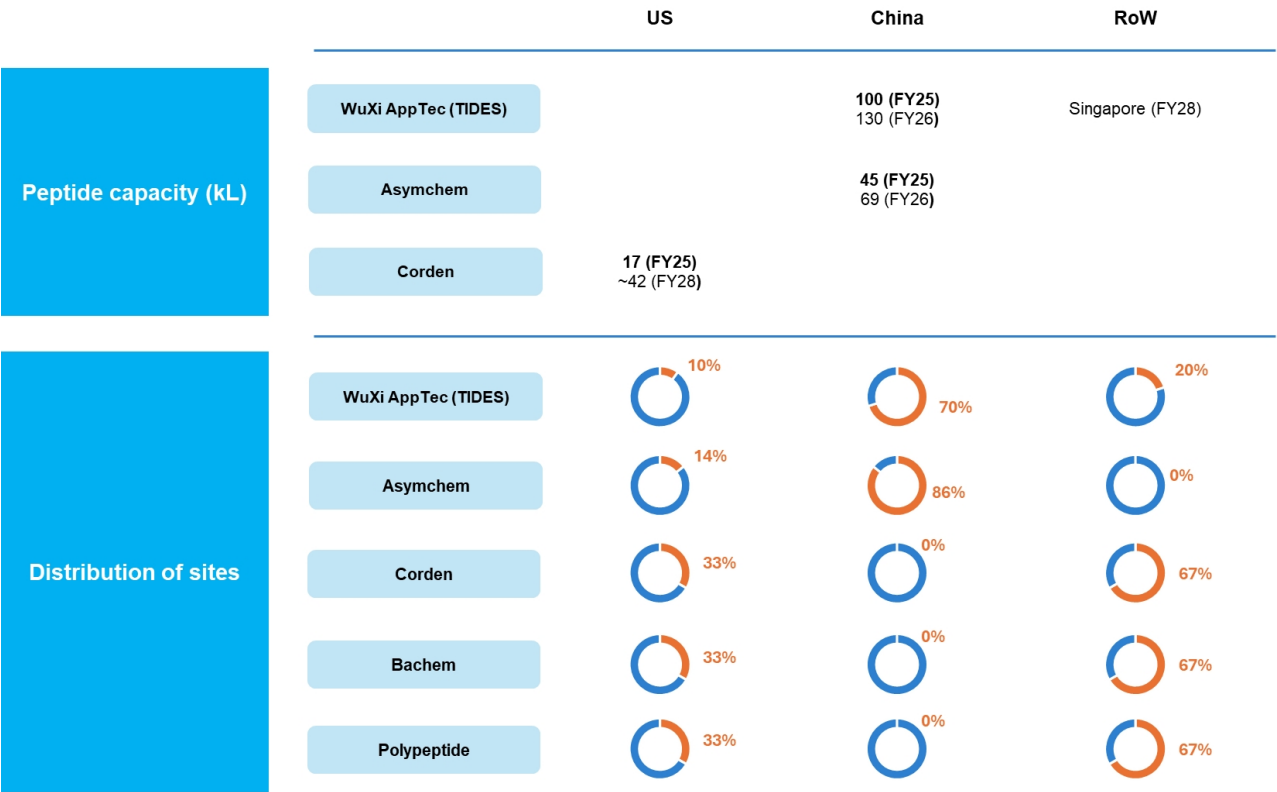
Note: Asymchem's annual peptide production capacity is estimated based on its reported reactor volume (kL)  
Source: Company disclosures, Bernstein analysis and estimates

EXHIBIT 7: Peptide manufacturing batch scale benchmark

| Company      | Batch size                                                                 |
|--------------|----------------------------------------------------------------------------|
| WuXi AppTec  | Estimated 25–30 kg average batch size <sup>1</sup> based on annual output. |
| Asymchem     | ~30 kg per batch (SPPS); >100 kg scale capability (LPPS).                  |
| CordenPharma | 12–10,000 L reactors; gram scale to 400 kg per batch.                      |
| Bachem       | Commercial peptide manufacturing with >15 kg batch capacity.               |
| PolyPeptide  | Not disclosed; estimated to be comparable to Bachem (>15 kg).              |

Note: 1. Based on 2000 batches and 58mt estimated peptide output in 2025.  
Max batch size not disclosed  
Source: Company disclosures, Bernstein analysis and estimates

EXHIBIT 8: Peptide manufacturing and development site landscape for global leading CDMOs



Source: Company disclosures, Bernstein analysis

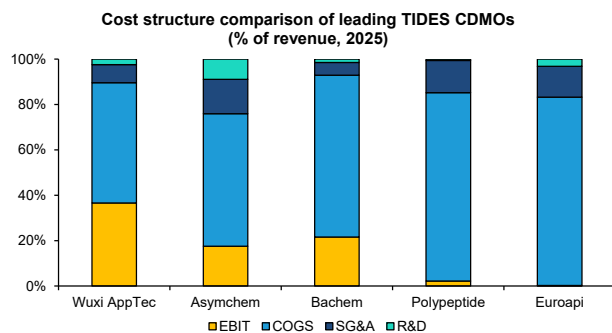
## Divergent cost structures

Despite operating within the same peptide value chain, profitability varies significantly across leading peptide CDMOs. WuXi AppTec stands out with a 47% GPM and 37% EBIT margin in 2025, supported by the lowest COGS ratio among peers at 53% of revenue. Of note, their group-level profit margin has improved over time with the increasing mix of TIDES revenue. By contrast, European specialty peptide players operate with materially lower profitability: Bachem generated c.22% EBIT margin, while PolyPeptide and EuroAPI remained near breakeven in 2025 (Exhibit 9).

**WuXi AppTec has been the industry's most aggressive capacity builder.** The biggest strategic differentiator among peptide CDMOs has been capital deployment. Since 2020, WuXi AppTec has consistently outspent all peers, from **\$460 Mn in 2020 to c.\$1.5 Bn in 2022**, before remaining elevated at **\$831 Mn in 2025**. In comparison, 2025 CapEx amounted to approximately **\$408 Mn for Bachem**, **\$191 Mn for Asymchem**, **\$124 Mn for PolyPeptide** and **\$89 Mn for EuroAPI** (Exhibit 12). This multi-year investment cycle positioned WuXi ahead of the GLP-1 demand wave and enabled the company to capture a disproportionate share of incremental outsourcing demand as metabolic peptide production ramped globally.

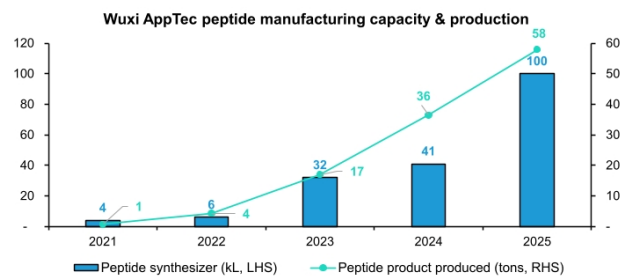
**ROIC varies significantly - Apptec has outperformed consistently.** CapEx intensity and returns on invested capital differ markedly across the peer group as well. Historically, Bachem and PolyPeptide operated with elevated **CapEx-to-sales ratios of 25-45%**, while WuXi AppTec's ratio peaked near **30% in 2021** before declining to roughly **12% in 2025** as revenue growth began to outpace incremental investment (Exhibit 13). More importantly, WuXi has demonstrated one of the strongest ROIC recoveries among peers, with ROIC expanding from the mid-teens in 2021 to approximately **38% in 2025** (Exhibit 14), driven by rapid utilization of newly built facilities (Exhibit 15). In contrast, several Western peers experienced declining or depressed ROICs during recent expansion cycles, underscoring the importance of filling capacity quickly once large-scale peptide plants come online.

**EXHIBIT 9: WuXi AppTec and Asymchem generate materially higher EBIT margins, while peptide specialists allocate a significantly larger proportion of revenue to manufacturing costs.**



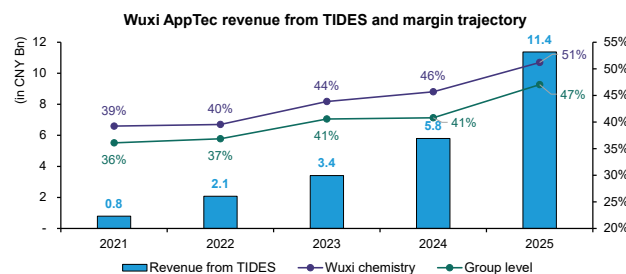
Source: Company disclosures, Bernstein analysis

**EXHIBIT 10: Peptide synthesizer capacity expansion has accelerated since 2023, driving a step-change in production for Wuxi AppTec**



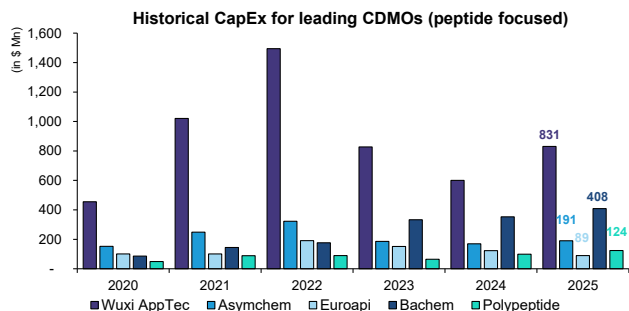
Source: Source: Company disclosures, Bernstein analysis

**EXHIBIT 11: WuXi AppTec TIDES: rapid revenue expansion driving margin improvement**



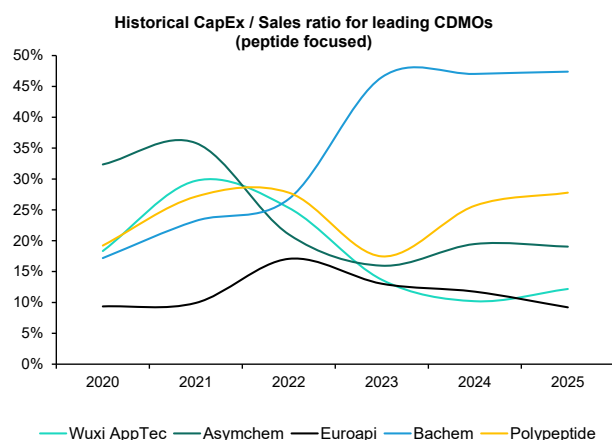
Source: Company disclosures, Bernstein analysis

**EXHIBIT 12: In absolute terms, WuXi AppTec has been the largest capital spender in the peptide CDMO space.....**

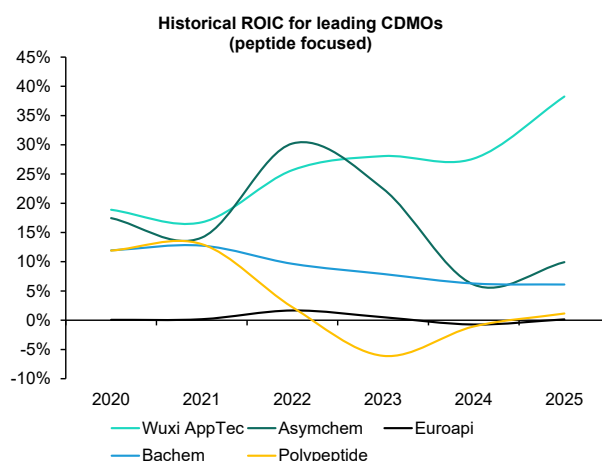


Source: Company disclosures, Bernstein analysis

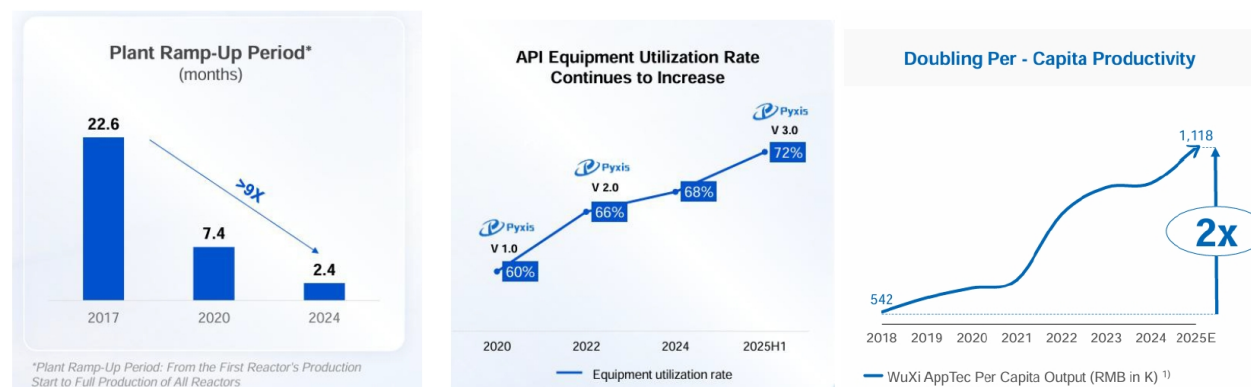


**EXHIBIT 13: .....while on a CapEx-to-sales basis, Bachem has been the most investment-intensive player**


Source: Company disclosures, Bernstein analysis

**EXHIBIT 14: WuXi AppTec leads peers in ROIC, demonstrating superior capital efficiency**


Source: Company disclosures, Bernstein analysis

**EXHIBIT 15: Apptec's higher ROIC has been driven by faster plant ramp-up and utilization / productivity gains**


Source: Wuxi Apptec Investor Day 2025

**Technology breadth and process efficiency**

**A variety of peptide modalities:** basic peptide modalities such as **cyclic peptides** and **hydrocarbon-stapled peptides** have become widely available across most leading peptide CDMOs. There is greater distinction in the breadth of **peptide conjugation**. While peptide conjugates with small molecules, lipids, carbohydrates, and proteins etc. are more commonly available, Wuxi Apptec and Asymchem have developed conjugates with newer modalities such as oligonucleotides and radionuclides. With a growing trend of multi-targeted and cross-modality approaches in drug research, we believe conjugation expertise to be an important differentiator beyond simple manufacturing scale (Exhibit 16-Exhibit 17).

**Materials and waste management and sustainability:** peptide manufacturing is a resource-intensive and highly pollutive process. Due to repeated coupling, deprotection, washing and purification steps, peptide synthesis can generate **process mass intensities (PMIs) that are roughly 50-100x higher than those of conventional small-molecule manufacturing**, resulting in significant solvent consumption and waste generation. As a result, leading suppliers have invested in greener chemistry, solvent recovery, process intensification and improved resin technologies to reduce waste. Among peers, we view **WuXi AppTec and Bachem as technology leaders in managing solvent and waste** (not surprisingly, also reflected in their higher GPM than peers), lowering PMI while simultaneously improving production efficiency at commercial scale (Exhibit 18).

**Faster development timelines are another advantage for Chinese players.** Speed-to-clinic and speed-to-commercialization have emerged as increasingly important considerations for peptide developers. Based on disclosed



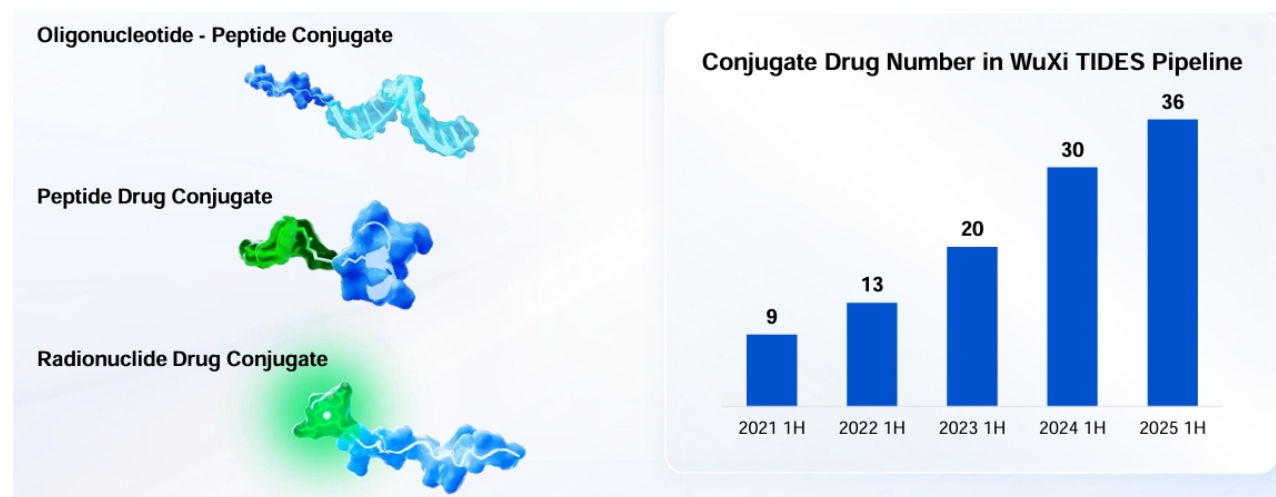
timelines, Chinese players generally deliver **shorter IND and NDA timelines** than Western peers (Exhibit 19).

EXHIBIT 16: **WuXi AppTec and Asymchem offer the broadest peptide technology portfolios**

| Company     | Peptide varieties |                               |                                     |                          | Peptide Conjugations |                        |          |                  |                |            |
|-------------|-------------------|-------------------------------|-------------------------------------|--------------------------|----------------------|------------------------|----------|------------------|----------------|------------|
|             | Cyclic peptides   | Hydro carbon stapled peptides | LNP formulation for Peptides/Oligos | CEPS & Synthetic Biology | Small molecule / PEG | Lipids / carbohydrates | Proteins | Oligo nucleotide | Radio nuclides | Antibodies |
| WuXi AppTec | ✓                 | ✓                             | ✓                                   | ✓                        | ✓                    | ✓                      | ✓        | ✓                | ✓              | ✓          |
| Asymchem    | ✓                 | ✓                             | ✓                                   | ✓                        | ✓                    | ✓                      | ✓        | ✓                | ✓              | ✓          |
| Corden      | ✓                 | ✓                             | ✓                                   |                          | ✓                    | ✓                      |          |                  |                |            |
| Bachem      | ✓                 | ✓                             |                                     |                          | ✓                    |                        | ✓        |                  |                |            |
| Polypeptide | ✓                 | ✓                             |                                     |                          |                      | ✓                      | ✓        |                  |                |            |

Source: Company disclosures, Bernstein analysis

EXHIBIT 17: **Conjugate drug modalities within TIDES and expansion of conjugate pipeline**



Source: Wuxi AppTec

## EXHIBIT 18: Peptide manufacturing sustainability benchmark

| Category / Company         | PMI <sup>1</sup> / sustainability metric           | Note                                                                                                                                                 |
|----------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Typical small molecule API | PMI: ~170–300                                      | Significantly lower material intensity                                                                                                               |
| Typical peptide API        | PMI: ~13,000–14,000                                | Peptide production is substantially more resource- and solvent-intensive                                                                             |
| WuXi AppTec (TIDES)        | Solvent optimization                               | Implemented solvent optimization and recycling initiatives, reducing total solvent use by 25% and replacing 50% of DMF with sustainable alternatives |
|                            | PMI reduced to <3,659                              | Achieved a <b>3–4x reduction in PMI</b> versus conventional peptide manufacturing processes                                                          |
| Bachem                     | Organic solvent usage reduced by up to 60%         | Leveraging green chemistry initiatives to improve process sustainability                                                                             |
|                            | >30% reduction in purification solvent consumption | Early adopter of continuous chromatography, improving purification efficiency                                                                        |

Note: 1. PMI (Process Mass Intensity) measures the total mass of raw materials, reagents, solvents, and water used per unit mass of API produced  
Source: Company disclosures, Bernstein analysis

## EXHIBIT 19: Peptide IND & NDA delivery timeline matrix for global leading CDMOs

| Company     | Peptide-to-IND timeline  | Peptide-to-NDA timeline |
|-------------|--------------------------|-------------------------|
| WuXi AppTec | 4 – 6 Months             | 12 – 15 Months          |
| Asymchem    | 3 – 6 Months             | 12 – 18 Months          |
| Corden      | ~11 Months (Includes DP) | 15 – 24 Months          |
| Bachem      | 6 – 9 Months             | 18 – 24 Months          |
| PolyPeptide | 6 – 9 Months             | 18 – 24 Months          |

Source: Company disclosures, Bernstein analysis

## PEPTIDE SYNTHESIS 101

### Synthetic vs. recombinant manufacturing

Peptides can be manufactured through either **chemical synthesis** or **recombinant biological processes**. Recombinant production uses engineered microorganisms or mammalian cells to express the peptide biologically, making it well suited for longer peptide chains and protein-like molecules without non-natural amino acids or chemical modifications. By contrast, **most commercially important peptide drugs today are manufactured chemically**, typically using solid-phase peptide synthesis (SPPS), which allows precise control over sequence modifications and conjugations (Exhibit 20-Exhibit 21).

### Semaglutide: a hybrid approach

Semaglutide represents one of the most commercially successful peptides ever developed, yet it is manufactured differently from most modern metabolic peptides. Novo Nordisk (covered by Justin Smith) uses a **hybrid process** in which the peptide backbone is first produced through recombinant expression before undergoing chemical modification and conjugation steps. A peptide precursor is generated biologically and subsequently linked to a fatty acid side chain that extends half-life and enables once-weekly dosing.

### Tirzepatide: a typical case of SPPS and LPPS manufacturing

Tirzepatide, the active ingredient in Mounjaro and Zepbound, exemplifies how most metabolic peptides are manufactured today. The molecule is produced through a largely **chemical process**, beginning with **solid-phase peptide synthesis (SPPS)**, in which amino acids are sequentially built onto a resin support one step at a time. For larger and more complex peptides, manufacturers often combine SPPS with **liquid-phase peptide synthesis (LPPS)**, where separately synthesized fragments are joined together before final conjugation and purification. In tirzepatide's case, four peptide fragments are synthesized, assembled and subsequently modified with its characteristic fatty acid chain, creating a highly scalable manufacturing route that is now widely adopted by leading peptide CDMOs (Exhibit 22-Exhibit 24).

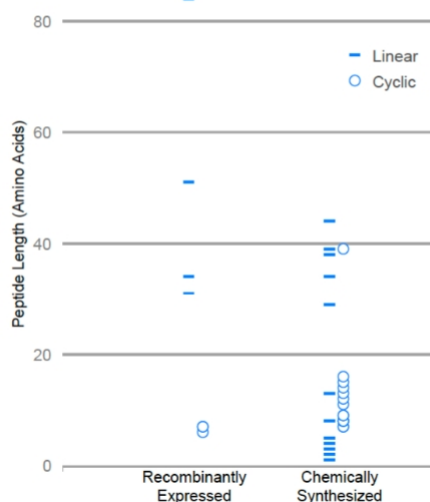
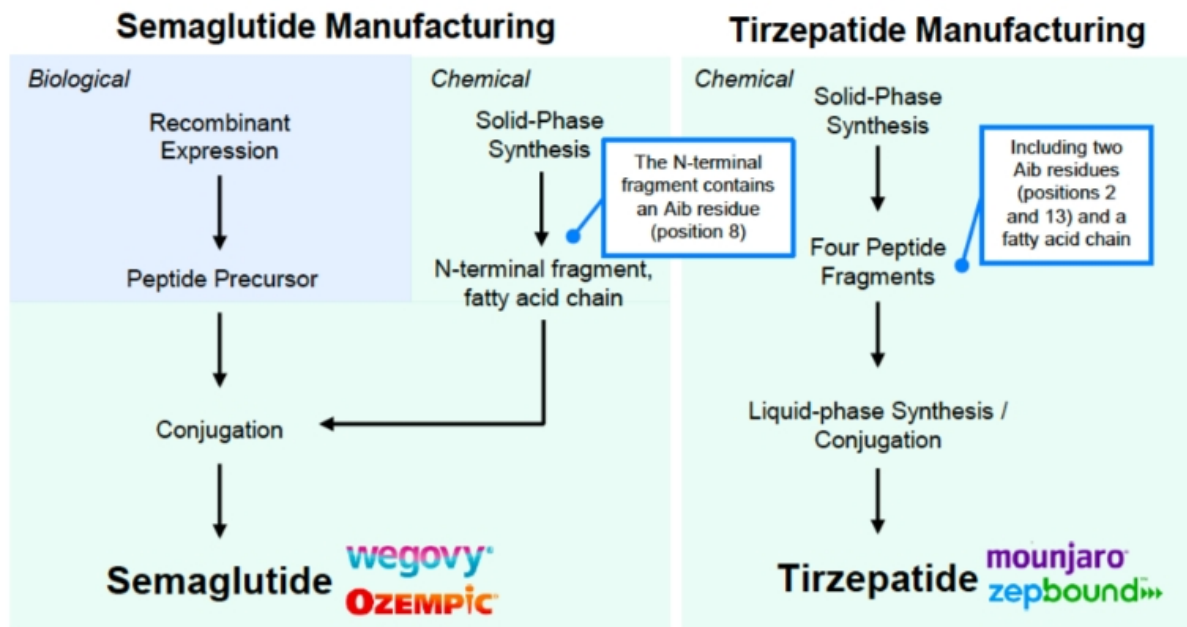
### Overview of the peptide manufacturing value chain

Peptide manufacturing begins with a relatively specialized supply chain consisting of amino acids, resins and coupling reagents, which serve as the fundamental building blocks for synthesis. These materials are converted into peptide intermediates through SPPS or LPPS, followed by fragment condensation and sequence modifications such as lipidation or stapling for more complex molecules. The resulting crude peptide must then undergo cleavage, deprotection, purification and lyophilization before becoming an active pharmaceutical ingredient (API). Finally, the API enters drug product manufacturing, where formulation, sterile fill-finish and packaging transform the molecule into the injectable or oral medicines ultimately delivered to patients (Exhibit 25).

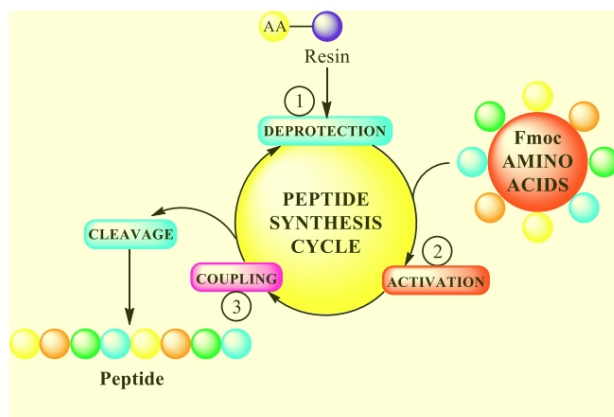
**EXHIBIT 20: Representative commercialized peptide drugs and their manufacturing - chemical synthesis is used more frequently than recombinant production**

| Brand Name          | Generic Name       | Drug Target                                             | Therapeutic Indications                                    | Primary Manufacturing Method                          |
|---------------------|--------------------|---------------------------------------------------------|------------------------------------------------------------|-------------------------------------------------------|
| Ozempic / Wegovy    | Semaglutide        | GLP-1R                                                  | Type 2 Diabetes, Obesity / Weight management               | Hybrid (Recombinant backbone + Chemical modification) |
| Mounjaro / Zepbound | Tirzepatide        | GIPR / GLP-1R (Dual agonist)                            | Type 2 Diabetes, Obesity / Weight management               | Chemical (SPPS / LPPS Hybrid)                         |
| Victoza / Saxenda   | Liraglutide        | GLP-1R                                                  | Type 2 Diabetes, Obesity / Weight management               | Hybrid (Recombinant backbone + Chemical modification) |
| Copaxone            | Glatiramer acetate | MHC Class II molecules (Immunomodulator)                | Relapsing Multiple Sclerosis (MS)                          | Chemical (Solution/Solid-phase polymerization)        |
| Tralicity           | Dulaglutide        | GLP-1R                                                  | Type 2 Diabetes                                            | Recombinant DNA (Mammalian cell culture)              |
| Sandostatin         | Octreotide         | SSTR (Somatostatin receptors, primarily SSTR2/5)        | Acromegaly, Severe diarrhea / Carcinoid syndrome           | Chemical (SPPS)                                       |
| Lupron / Eligard    | Leuprolide         | GnRHk (Gonadotropin-releasing hormone receptor agonist) | Prostate Cancer, Endometriosis, Central Precocious Puberty | Chemical (SPPS)                                       |
| Forsteo / Forteo    | Teriparatide       | PTH1R (Parathyroid hormone receptor type 1)             | Severe Osteoporosis                                        | Recombinant DNA ( <i>E. coli</i> expression)          |
| Zoladex             | Goserelin          | GnRHR                                                   | Prostate Cancer, Breast Cancer, Endometriosis              | Chemical (SPPS)                                       |
| Somatuline          | Lanreotide         | SSTR                                                    | Acromegaly, Neuroendocrine Tumors (NETs)                   | Chemical (SPPS)                                       |

Source: company disclosure, Bernstein analysis

**EXHIBIT 21: Chemical synthesis is the far more common route for recently approved peptide drugs: 31 out of 38**
Source: <https://www.decibio.com/insights/chemical-vs-recombinant-the-manufacturing-decisions-behind-blockbuster-peptide-drugs>
**EXHIBIT 22: Differences between manufacturing routes of semaglutide (recombinant followed by chemical conjugation) and tirzepatide (chemical synthesis)**
Source: <https://www.decibio.com/insights/chemical-vs-recombinant-the-manufacturing-decisions-behind-blockbuster-peptide-drugs>

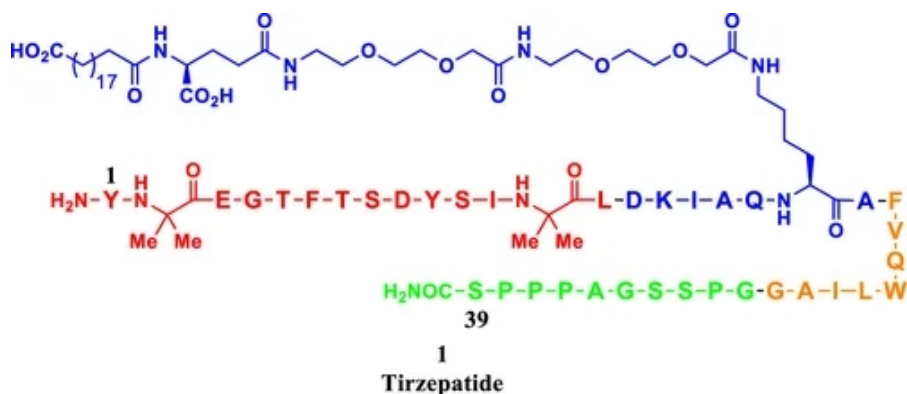
**EXHIBIT 23: Solid phase peptide synthesis (SPPS) involves an iterative process of deprotection, amino acid activation, and coupling to assemble amino acids into peptide chains**



**Figure 2.** General SPPS methodology for the synthesis of peptides via (1) deprotection; (2) amino acid activation; and (3) coupling. Note: an optional capping step is often included in a SPPS process but is not included in the figure.

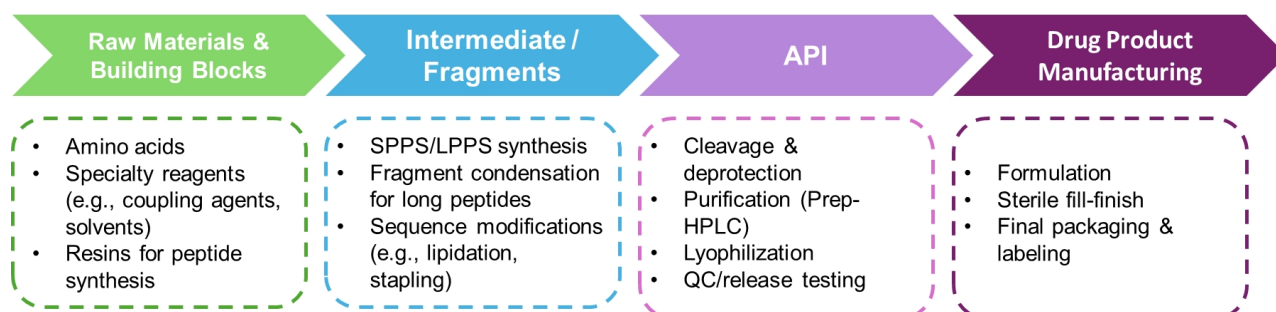
Source: <https://pubs.acs.org/doi/10.1021/acs.oprd.1c00108>

**EXHIBIT 24: Tirzepatide is made by four fragments synthesized through SPPS and joined to make the final product through LPPS**



Source: <https://pubs.acs.org/doi/10.1021/acs.oprd.1c00108>

EXHIBIT 25: **Overview of the peptide manufacturing value chain**



Source: Company disclosures, Bernstein analysis

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We use a blended approach combining a DCF valuation (CNY198, applying a WACC of 11.0% and a 2.5% terminal growth rate) with 33% weight, P/E multiples (CNY204, using a 1-year forward P/E ratio of 40x) with 33% weight, and EV/EBITDA (CNY186, using a 1-year forward EV/EBITDA ratio of 24x) with 33% weight, leading to a one-year price target of CNY198. H share TP is calculated by applying 2026 YTD stock price premium 0.79, leading to one-year price target of HK\$ 157.

**WuXi AppTec Co Ltd**

We use a blended approach combining a DCF valuation (CNY162, applying a WACC of 10.6% and a 2.5% terminal growth rate) with 33% weight, P/E multiples (CNY177, using a 1-year forward P/E ratio of 25x) with 33% weight, and EV/EBITDA (CNY166, using a 1-year forward EV/EBITDA ratio of 17x) with 33% weight, leading to a one-year price target of CNY168. H share TP is calculated by applying 2026 YTD stock price premium 1.21, leading to one-year price target of HK\$ 204.

**Wuxi Biologics Cayman Inc**

We use a blended approach combining a DCF valuation (CNY28, applying a WACC of 10.9% and a 2.5% terminal growth rate) with 33% weight, P/E multiples (CNY43, using a 1-year forward P/E ratio of 25x) with 33% weight, and EV/EBITDA (CNY47, using a 1-year forward EV/EBITDA ratio of 17x) with 33% weight, leading to a one-year price target of CNY39, equivalent to HK\$42.

**RISKS****Asymchem Laboratories Tianjin Co Ltd**

Downside risks: 1) Service quality deteriorates and higher drop-out rate for integrated projects; 2) Geo-political tension escalates, resulting losing share in Global-ex-China market; 3) New treatment modalities like peptides have higher failure rate than we estimated

**WuXi AppTec Co Ltd**

Downside risks: 1) Service quality deteriorates and higher drop-out rate for integrated projects; 2) Geo-political tension escalates, resulting losing share in Global-ex-China market; 3) New treatment modalities like peptides have higher failure rate than we estimated

**Wuxi Biologics Cayman Inc**

Upside risks: 1) Better than expected performance of biologics and ADC business; 2) Large commercial order obtained

Downside risks: 1) Service quality deteriorates and higher drop-out rate for integrated projects; 2) Geo-political tension escalates, resulting losing share in Global-ex-China market; 3) New treatment modalities like ADC and BsAb have higher failure rate than we estimated

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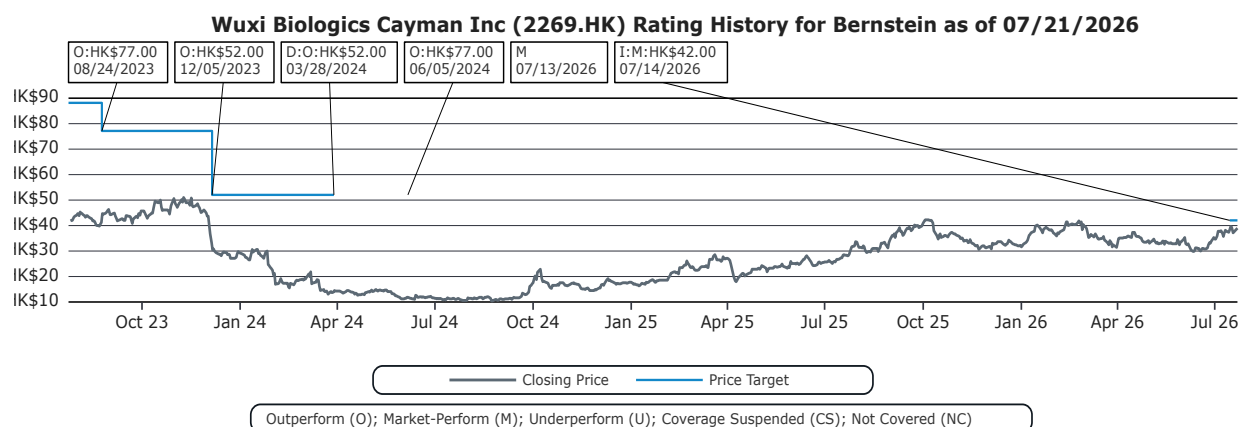
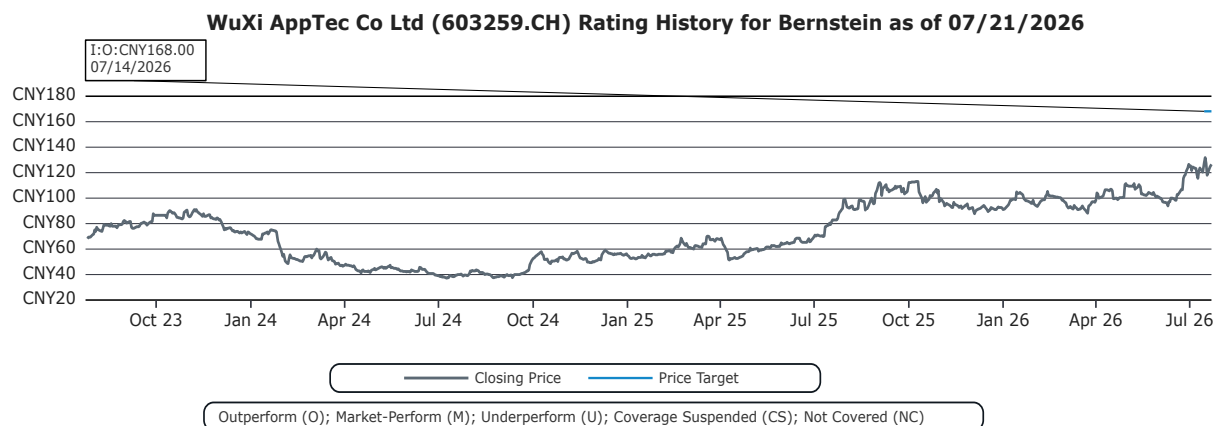
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